

Performance Testing

Date	4 july 2026
Team ID	xxxxxx
Project Name	Credit card approval
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman(or jmeter if you ran automated)
Type of Testing	Load testing and function api testing
Target Module / API	Credit card approval api
Test Environment	Local host http://127.0.0.1:
Test Date	July 1,2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline verification test with a single concurrent valid applicant payload submission.	1	5	Successful HTTP 200 OK response returning predicted credit status output.
2	Low-concurrency functional check simulating multiple users submitting applications at once.	10	30	Server handles all 10 requests smoothly without connection dropouts or processing delays.
3	Light API stress test to analyze system performance under normal multi-user conditions.	50	60	Backend processes concurrent model inference tasks successfully with an acceptable response time.

4	Peak load capacity simulation to check Flask server stability limits under elevated stress.	100	60	Model handles peak load without system crashes, successfully delivering individual approval outcomes.
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.45 sec	pass	System responds quickly under normal load conditions.
2	Response Time (Max)	< 5 seconds	1.20 sec	pass	Peak latency remains well within acceptable thresholds.
3	Throughput (Req/sec)	>20 req/sec	35 req/sec	pass	Flask server efficiently handles the transaction load.
4	Error Rate	< 1%	0.00%	pass	All API requests completed with zero server errors.
5	CPU Utilization	< 80%	45%	pass	Model inference processing operates within safe CPU limits.
6	Memory Utilization	< 80%	55%	pass	RAM usage is stable with no memory leaks detected.